Hidden counting phases obstruct exact displayed-state Markov closure

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

PAH-STATE-001 asks whether the source display $d(x) = (s, \psi, U)$ is a sufficient state for an exact time-homogeneous Markov description of the unchanged PAH-v2 counting process. We disprove this at one already-admitted finite instance, including its original stationary Gibbs law. We do not reject the full counting process. The preregistration, source locators and hashes are in strategy/pa-hyp/PAH-STATE-001-prereg-v1.json; its SHA-256 is

5d88addeb96c8cc4a95164a0f92da227b0baf53275be7081c4ea78f18f7a2943

The original functional is PAH-001-v1, incorporated unchanged by PAH-001-v2-r2. Counting tuples, all signed PH/TR/LK/AP roots, P_{cand}, mobility, Gibbs normalization and external stochastic time are unchanged. Conditional expectation below is a diagnostic in the original probability space, not a new dynamics or physical projection.

Literature and prior scope. Tian and Kannan, *Lumpability and Commutativity of Markov Processes*, Stochastic Analysis and Applications 24 (2006), 685–702, DOI 10.1080/07362990600632045, Definition 2.9 and Theorems 2.10–2.11, p. 689, give the finite generator criterion. Finite state, fixed partition and homogeneous generator hypotheses hold here; the fibre-rate condition is tested, not assumed. We rederive the necessary identity below and the stationary compression argument directly. This is standard finite Markov theory applied to a model-specific state defect, not a new general theorem. The older free-vertex restriction (EXP-001362) forgets spatial sites; this test forgets only coincident-value labels on the same carrier. R-570 is used only for the existing finite model, with detailed balance rechecked. R-575 and R-576 concern cutoff freezing and stay intact.

2. Exact admitted witness

Use the adopted Morton square $G_{0,0}$, width 2, vertices $0 = (0,0), 1 = (1,0), 2 = (0,1), 3 = (1,1)$, edges $01, 02, 13, 23$, and the one closed square face. The anchors are $O = \{0\}$ and $C = \{1\}$; no field is clamped. This is the original open boundary. Set $r = h = N = 0$ and $m_0 = q_0 = R_0 = 2$. The source schedule gives $K = 2, M_s = 1, M_\psi = 2, Q = 2, R_{\max} = 2$. Take $\epsilon = 1/2$, $\beta = \nu = 1$, all positive functional couplings equal to 1, and $m^2 = \lambda_4 = 0$. These are admissible witness inputs, not a fit or a new source law. Every parameter is fixed; no limit is taken.

Consider the four counting states

$$j = (1,1,1,1), \quad \ell = (0,1,1,0), \quad n = (a,0,0,b), \quad u = (0,0,0,0), \qquad a, b \in \{0,1\}.$$

They have identical display $s_v = 1$, $\psi = (0,1,1,0)$, $U_e = 1$. This is the whole exact-display fibre: positive amplitudes fix n_1, n_2, zero amplitudes leave n_0, n_3 free, and $\epsilon < 1$ fixes every j. Use $f = |\psi_O|^2 = \ell_0^2$, a

gauge- and anchor-invariant member of the existing full finite algebra. It is not a replacement observable domain.

Each initial energy is $F = 4/3 + 2 = 10/3$: the first term is onsite and the second is the covariant gradient contribution. PH, LK and AP leave f unchanged. Of the four valid TR roots, only $1 \to 0$ and $2 \to 0$ change f; each increases it by one. Their mobility is one. TR retains all phase labels, including a at the initially empty vertex. The two target energies are therefore

$$F_{\text{target}} = \begin{cases} 4/3 + 1 = 7/3, & a = 0, \\ 4/3 + 3 = 13/3, & a = 1. \end{cases}$$

Consequently the original midpoint rate gives, exactly,

$$(Lf)(x_{0b}) = 2e^{1/2}, \qquad (Lf)(x_{1b}) = 2e^{-1/2}. \tag{1}$$

Each initial state has PH:8, TR:4, LK:8, AP:4 valid labelled incidences. K=2 duplicate signs are retained. No invalid root or root multiplicity is silently changed. As full-functional regression controls, the AP-minus energy increments at empty O and occupied C are $17/24$ and $1/3$; their mobility is $1/\sqrt{2}$, but their f increments are zero.

If a generator $\bar{L}$ on displays satisfied $L(\bar{f} \circ d) = (\bar{L}\bar{f}) \circ d$ for every displayed observable, its right side would be constant on a display fibre. Equation (1) contradicts this. The universal displayed-state closure route is therefore false at this admitted instance. This is not an all-regulator assertion.

3. Original stationary Gibbs law does not remove the defect

Let $\pi(x) = Z^{-1}e^{-F(x)}$ on the complete finite counting space. For each valid signed incidence, inverse validity and symmetric mobility give

$$\pi(x)c_r(x) = Z^{-1}m_r(x)e^{-[F(x)+F(rx)]/2} = \pi(rx)c_{r^{-1}}(rx).$$

Summation preserves all duplicate labels. Hence $L = L^*$ in $L^2(\pi)$, $S_t = e^{tL} = S_t^*$, and π has full support. Let $P = \mathbb{E}_\pi[\cdot \mid d]$, the orthogonal projection onto display functions, solely for this proof. Under the unchanged stationary path law, the displayed two-time conditional transitions are $C_t = PS_tP$ on $\operatorname{ran} P$. An autonomous stationary Markov description would require $C_{2t} = C_t^2$.

Finite matrix calculus and orthogonality instead yield the exact identity

$$C_{2t} - C_t^2 = PS_t(I - P)S_tP, \qquad \langle f, (C_{2t} - C_t^2)f\rangle_\pi = \|(I - P)S_tf\|_\pi^2. \tag{2}$$

All four fibre states have equal Gibbs weight. From (1),

$$\operatorname{Var}_\pi(Lf \mid d = d(x_{00})) = (e^{1/2} - e^{-1/2})^2 > 0.$$

Their total mass is $4e^{-10/3}/Z > 0$, so

$$\|(I - P)Lf\|_\pi^2 \geq \frac{4e^{-10/3}}{Z}(e^{1/2} - e^{-1/2})^2 > 0. \tag{3}$$

Since $Pf = f$ and the finite matrix exponential is analytic, $(I - P)S_tf = t(I - P)Lf + O(t^2)$ in the fixed finite norm. Equations (2)–(3) give a positive Chapman–Kolmogorov defect for all sufficiently small $t > 0$. Thus the actual stationary displayed process is not a time-homogeneous Markov chain. Merely averaging instantaneous rates over the Gibbs fibre cannot reproduce its finite-time law. There is no quantitative cutoff-uniform lower bound in (3), no change of initial state, and no interchange of time or regulator limits.

Gauge quotient. This is not a gauge artefact. Preserving all displayed $U = 1$ on the connected square forces a constant gauge label, and the two occupied phases then force that label to zero. The four hidden states are not identified by this stabilizer. The gauge orbit has 16 exact-display representatives, each with the same four-state conditional split because F and Lf are invariant. The conditional variance therefore persists also on the gauge-orbit display fibre. The two adjacent marked anchors make the cell automorphism group trivial. This additional observation does not assert that every arbitrary further quotient is non-Markov.

4. Verification and Devil's-advocate review

The primary script derives the regulator, source-root rows, full energies, balance, gauge action and fibre variance. A non-importing independent Fraction-based evaluator identifies the fibre by checking all 40,960 counting states; this enumeration checks a finite definition, not a limit. Independent analytic and hostile in-session reviews check (2)–(3). The hostile runner tests semantic corruptions and optimized-Python rejection. Four pinned Lean declarations check the abstract nonfactorization implication, positive exponential gap, variance algebra and positive-mass contribution. Lean does not encode the source functional, conditional expectation or full semigroup argument. See the hash-bound review and run JSONs for executed dispositions. No external-person certification is claimed.

- **Sign and factor (DISMISSED).** Lower target energy gives the larger rate; the exponent is $-\Delta F/2$. Both contributing TR roots remain, as do zero-increment PH/LK/AP roots.
- **Gauge and hidden-label redundancy (DISMISSED).** The witness is $|\psi_O|^2$, not a gauge phase. The exact display stabilizer and invariant rate calculation exclude a gauge-copy explanation.
- **Only artificial starts fail (DISMISSED).** The finite positive Gibbs fibre and reversible compression identity prove the stationary defect, without replacing the state by a conditional model.
- **Wrong carrier (VALID, corrected before preregistration).** A preliminary 3-by-3 sketch was rejected: the adopted widths are powers of two. All issued calculations use the actual $G_{0,0}$ and its anchors.
- **Limits and physical interpretation (UPHELD boundary).** The fibre mass need not stay bounded below in any limit. Neither this result nor formal verification identifies hidden labels with nature.

Independent external review of the exact source crosswalk and stationary compression proof is invited. Reproduce the complete checkpoint with

```
python -X utf8 verification/scripts/pah_state001_verify.py
  --lean-cache E:/Dev/TECT/verification/lean/.lake/packages
```

Run these two displayed lines as one command.

5. Consequence and Result footer

An exact memoryless realization of the full source model cannot use just these displayed fields as its state. A future physical packet must address latent state information or non-Markov memory and justify its interactions; this theorem neither supplies such a packet nor identifies a minimal additional state. OBS-002 remains parked. R-570, R-575 and R-576 are preserved. No model, state, rate, clock, comparison map or limit is repaired. No automatic successor is authorized by this result.

Result ID: R-577 (C6-SPACETIME-SIGNATURE)
Precise statement: Exact displayed-state Markov closure fails.
Scope: One fixed admitted PAH-v2 finite instance, original Gibbs.
Dependencies: Frozen PAH-v1/v2 definitions; finite reversibility.
Evidence grade: ANALYTIC, EXACT, EXECUTED, INHERITED.
Reproduction command: python verification/scripts/pah_state001_verify.py
Expected output: PASS exact witness and partial Lean checks.
Falsification gate: A wrong source/root/fibre or compression identity.
Tier before / after: C6 T1 / T1; no mainline gate change.
No-overclaim statement: No physical Pre-A, QFT, gravity or limit.
Next required action: Park; require physical latent-state/memory evidence.